ARCHIE BENN

HOW TO TRAIN YOUR MODEL

ASSESSING THE IMPACTS OF FOREST AGE ON THE SPATIAL AND TEMPORAL GENERALISATION OF RANDOM FOREST EVAPOTRANSPIRATION MODELS

MSc Bioinformatics
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ABSTRACT

The ability to accurately predict actual evapotranspiration (ET) in regions using methods requiring minimal data input would enable estimations to be made in areas which are underrepresented globally due to the high costs involved in direct ET measurements. These estimations could, in turn, aid in more efficient use of freshwater and in predicting natural phenomena such as droughts. The development of ML models to predict ET has gained popularity in recent years, and being able to understand what links exist between a training dataset and a model which performs well on out-of-sample data should aid in the formation of better-predicting models, thus any explanatory variables which may aid in ET estimations should be explored. One under explored feature in ET ML models is forest age, which may impact ET rates due to age-related changes to canopy complexity and hydraulic conductance. The present study considers the use of forest age as a feature in ML predictions of ET at forest sites contained within the FLUXNET by using methods to explore spatial generalisation of models trained with and without age included, alongside temporal generalisation to assess whether forest age may help predictions across time at single sites. Results from spatial generalisation suggest nuanced results, with age benefiting predictions at some held out test sites, and hurting at others. Further investigation into two sites with opposing results pointed towards x being a factor (MAYBE?). Temporal generalisation suggested forest age x y z. These findings suggest that forest age may be useful to consider in ML ET modelling in the future, however considerations relating to tree species and the impacts of age on tree dynamics within each species may be useful in determining whether to use forest age or not in predictions. Suggestions for future work include accurate age surveys to be carried out at other FLUXNET sites which may permit 'per-species models' to be formed and tested for use with forest age.

Key words: Machine Learning, Evapotranspiration, FLUXNET, Forest Age.
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Introduction: 242
Results: 2471
Discussion: 861
Conclusion: 1

DEDICATION AND ACKNOWLEDGMENTS

I would like to thank my supervisor Martin de Kauwe for his help and guidance throughout the duration of this project.

DECLARATION

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Archie Benn
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CONTENTS

1 AIMS AND OBJECTIVES

The overarching aim of this research project is to investigate how the age of a forest may impact predictions of actual evapotranspiration rates (hereafter ET) using machine learning (ML) models in forest ecosystems. The stand age of a forest has been shown to impact ET dynamics [1], and therefore could alter how ML models form predictions of ET if age is included as a feature during training and inference.

Given the importance of predicting ET [2, 3] and the lack of infrastructure to measure ET directly with eddy covariance (EC) methods in many ecoregions [4], the first part of this project will focus on spatial generalisation, asking: *"If a model is built using readily available data from one set of sites, how well can it form predictions of ET rates at a geographically unseen site, and how does utilising forest age affect predictions?"*. To address this question, FLUXNET sites with forests aged by Besnard et al. [5] are used alongside a framework developed to quantify site similarities based on characteristics in section 2.1. Random Forest (RF) models were trained with and without forest age as a feature, and predictions of ET rates from both directly compared at unseen sites with ground-truth validation provided by EC measurements.

The objectives of this project are twofold: (1) To assess the use of forest age in models for spatial predictions of ET, and (2) to outline future directions for research in utilising forest age in ET predictions.

2 RESULTS

2.1 Flux Tower Site Clustering

Cluster analysis was carried out in order to classify sites into groups based on their characteristics. This classification permits the predictability of ET, both within these characterised groups and across them, to be investigated. Gower's distances, a measure between pairs of data points, from 0 = completely dissimilar to 1 = completely similar [6] (see section 6.2 for details), were computed using the features listed in Table 4 and saved as a matrix. This matrix was passed to the Partitioning Around Medoids (PAM) algorithm [7] with the silhouette method to form clusters and identify medoid sites - those which have minimal dissimilarity to others within each cluster and thus can be seen as the representative sites of each cluster [8].

The silhouette method identified $k = 13$ as the optimal number of clusters based on average silhouette width, but with only 50 sites present this value was dropped to $k = 6$ in order to yield sufficiently well-populated clusters across a range of site characteristics. The cluster assignment solution from PAM with $k = 6$ clusters resulted in a range of site counts by cluster from 3 sites (cluster 6) up to 18 sites (cluster 3), and can be seen in a cluster plot of the first two principal components (Figure 1a). The overall geographic distribution of clusters appears to follow an expected pattern with regards to climatic groups; for example, sites at high latitudes tended to be grouped in cluster 3, with Mediterranean and coastal sites typically grouped in cluster 5 (Figure 1b). This expected pattern of clusters suggests that the PAM clustering solution captures climate-relevant patterns to some degree, although this was not formally tested. The six medoid sites were geographically distributed with four sites in North America and two sites in Europe (Figure 1b), and a range of site ages from 20 years (US-Me3) to 117 years (DE-Lnf).

To define cluster site characteristics, the means of z-score standardised numerical features (mean = 0, sd = 1) and mode of categorical features were considered for each cluster. Mean scaled numeric features can be seen within a heat map (Figure 2) for a visual comparison across clusters. Mean cluster features ≥ 1 sd from the all site mean were considered as 'high/low' (≥ 1.5 sd as 'very high/very low') in order to qualitatively classify cluster characteristics, with these summarised in Table 1.

2.2 Model Sensitivity

Results from model sensitivity testing indicate mixed model performance across the medoid sites, based on regression metrics comparing the predicted and observed daily ET Figure 3. In this part of the analysis, features were sequentially added to RF models, and these were then re-trained with each additional feature before testing on all medoid sites (see section 6.3.1 for details).

Predictions of daily ET formed for CA-Obs, US-UMd, and DE-Lnf showed strong performance across the range of input features, while CA-TP3, IT-SRo, and US-Me3 performed poorly when considering R^2, RMSE and MAE.

CA-Obs and DE-Lnf showed very similar patterns of performance, with predictions at both sites typically strong and climbing across metrics up until `cos_DOY` was added (Peak CA-Obs: R^2 = 0.90, RMSE = 0.28, MAE = 0.19; Peak DE-Lnf: R^2 = 0.90, RMSE = 0.38, MAE = 0.28). Predictions at both sites then declined with the inclusion of `Site_age` to the model (ΔR^2 from `cos_DOY` → `Site_age`: CA-Obs = -0.07; DE-Lnf = -0.04). US-UMd deviated from this performance pattern slightly, with metrics improving further when `Site_age` was included in the model (Peak US-UMd: R^2 = 0.90, RMSE = 0.41, MAE = 0.28, ΔR^2 from `cos_DOY` → `Site_age` = +0.03). At all three of these sites the inclusion of only a single feature in the RF model, `T_air`, demonstrated strong to moderate predictive performance (CA-Obs: R^2 = 0.76, RMSE = 0.42, MAE = 0.31; US-UMd: R^2 = 0.64, RMSE = 0.78, MAE = 0.58; DE-Lnf: R^2 = 0.63, RMSE = 0.73, MAE = 0.58).

Predictive performance for daily ET at IT-SRo generally improved up until a peak with the addition of `Lai_500m` before falling again with the addition of later variables, but overall predictive performance was poor at this site (Peak IT-SRo: R^2 = 0.28, RMSE = 0.92, MAE = 0.67). Performance at the remaining two sites, CA-TP3 and US-Me3, was particularly mixed and relatively poor (Peak CA-TP3: R^2 = 0.55, RMSE = 0.70, MAE = 0.48; Peak US-Me3: R^2 = 0.00, RMSE = 0.75, MAE = 0.56). R^2 fluctuated between predictions made with consecutively added features at these two sites, with performance at US-Me3 indicating that the model has no explanatory power at this site based on an R^2 maximum of 0.0.

Ratios of RMSE:MAE from peak predictive models at each medoid all fall between 1.33 and 1.48, suggesting that the overall impact of outlier ET predictions, and therefore overall error distribution, was fairly consistent from the best performing models at all sites during this analysis.

2.3 Train-Test Distance Analysis From Sampled Training

Figure 4 shows the paired results between the 'Mod' and the 'Mod + Age' models (described by Equation 1 and Equation 2 respectively) during the shuffled and sampled training set analysis (see section 6.3.2 for details).

Results from this analysis indicate that including age had a significant impact across a range of training set site compositions, with all six medoid sites showing significant differences in R^2 between the 'Mod' and the 'Mod + Age' predictions for daily ET (Wilcoxon signed-rank tests, $p < 0.01$ for all). The largest average effect on R^2 between the two models was at US-Me3, with 58/100 runs resulting in a decreased R^2 when age was added (mean ΔR^2 = -0.10). Predictions at US-Me3 were poor throughout and also had a very high spread of R^2 variability which further increased when age was added (R^2_{Mod} = 0.04 ± 0.24; $R^2_{Mod+Age}$ = -0.06 ± 0.40). The predictions at CA-TP3 saw the greatest improvement to R^2 when age was added, where 75/100 runs increased R^2 (mean ΔR^2 = +0.02), although the spread of variability also increased alongside this (R^2_{Mod} = 0.53 ± 0.05; $R^2_{Mod+Age}$ = 0.55 ± 0.08).

Model predictions made at DE-Lnf and US-UMd were strong in both models, although opposite effects were observed at these sites during these runs. DE-Lnf predictive performance was negatively impacted with the addition of age (mean ΔR^2 = -0.05), with R^2 dropping during 88/100 runs and the spread of variability increasing nearly four-fold (R^2_{Mod} = 0.89 ± 0.013; $R^2_{Mod+Age}$ = 0.84 ± 0.050). US-UMd performance increased modestly (mean ΔR^2 = +0.02) and increased during 85/100 runs, with the spread of variability in R^2 remaining consistent between the models (R^2_{Mod} = 0.86 ± 0.02; $R^2_{Mod+Age}$ = 0.88 ± 0.02).

To investigate how the composition of the training dataset impacted model predictions of ET, values of the mean Gower's distance and mean age distance between training and test sites were formed during each of the 100 runs (see section 6.3.2 for details).

A visualisation of these results in Figure 5 for the full set of sites follow an expected pattern with predictive accuracy falling with increased Gower's and age distances across both models, implying that as the training set of a model becomes less similar to the test site, predictions tend to become less accurate. Multiple linear regression models were used to see if mean Gower's distance or mean age distance significantly predict R^2 in both models. The overall regression for 'Mod' was statistically significant ($R^2 = 0.60, F(2, 597) = 453.8, p < 0.001$), and both variables independently predicted R^2_{Mod} (Gower's: $\beta = -6.19, p < 0.001$; Age: $\beta = -0.0064, p < 0.001$). The overall regression for 'Mod + Age' was statistically significant ($R^2 = 0.54, F(2, 597) = 349, p < 0.001$), and both variables independently predicted $R^2_{Mod+Age}$ (Gower's: $\beta = -7.03, p < 0.001$; Age: $\beta = -0.0056, p < 0.001$).

As site age was included as one of the variables during cluster analysis (Table 4), mean Gower's distance and mean age distance are not fully independent ($VIF = 1.52$ for both predictors in both multiple linear regression models), however low multicollinearity indicates each predictor contributes distinct information within the models.

Across all sites together, greater dissimilarity between Gower's distances and age distances between the test site and training set sites is significantly associated with a lower R^2 in both models. However, this relationship is not present within every site individually. Site-specific simple linear models suggest Gower's distance showed no significant relationship with R^2 at CA-TP3 in either model, and was significantly positively associated with R^2 in 'Mod + Age' predictions at US-Me3. Similarly, mean age distance was only significantly associated with R^2 at US-UMd for both models, with no significant relationship present at the other five medoid sites in either model.

2.4 Case Study of US-UMd and DE-Lnf

US-UMd and DE-Lnf share similar site qualities (Table 7) and both show strong R^2 for daily ET predictions in both models, yet the predictive performance responses in both section 2.2 and section 2.3 were opposite when age was added as a feature to the models (for example, in Figure 4). New RF models were trained and tested using a single-site cross-validation approach (section 2.4.1), SHAP feature performance was evaluated (section 2.4.2 and 2.4.3), and training dataset ages were manipulated and ET predictions reevaluated (section 2.4.4). See section 6.3.3 for full details.

2.4.1 Observed vs. Predicted

Figure 6a and Figure 6b display scatter plots comparing the predicted and observed daily ET from each model at the sites. Visual comparisons of these plots suggests a tightening of points relative to the 1:1 line with the addition of age at US-UMd, particularly at the extremes. At DE-Lnf, the addition of age appears to cause low observed ET to be over-predicted, particularly in winter and autumn, in an opposite response to predictions of low observed ET at US-UMd. Both models at both sites appear to under-predict peak daily ET in summer given the clustering of points below the 1:1 line at higher observed ET. Performance results from regression metrics followed the same pattern as in sections 2.2 and 2.3, with R^2 improving in US-UMd predictions (R^2_{Mod} = 0.86; $R^2_{Mod+Age}$ = 0.90) and dropping in DE-Lnf predictions (R^2_{Mod} = 0.90; $R^2_{Mod+Age}$ = 0.86) with the addition of forest age as a feature.

2.4.2 Time Series Plots and SHAP Feature Importance

Time series plots of predictions from 'Mod + Age' (Figure 7a and Figure 7b) indicate strong cyclical predictive performance overall at both sites. Both models captured abnormal lows during summer months; for example, at US-UMd in summer 2011 largely captured an ET rate which is typically seen during mid-winter, demonstrating the flexibility of these models. Despite this, extreme ET peaks were missed in consecutive years at US-UMd; for example, a peak in summer 2010 was under-predicted by the model by nearly 2mm/day. Winters appeared to be problematic for the models, with ET estimations being consistently over-predicted and troughs missed in all years at both sites. SHAP feature importance order was similar at both sites, with `SW_rad` and `cos_DOY` dominating predictions in both 'Mod + Age' models overall, followed by `T_air` and `LAI`. Site age was ranked as the 6th most important predictor at US-UMd and the 5th most important at DE-Lnf. Mean SHAP values also indicate that site age had opposite impacts on predictions at both sites: age typically reduced daily ET predictions at US-UMd and raised them at DE-Lnf.

2.4.3 SHAP heatmap and ΔPredicted ET

To assess how SHAP values changed over time in the 'Mod + Age' model, a heatmap plot of mean weekly SHAP importance over time for features at both sites was formed. These were plotted below a graph which represents how daily ET predictions changed (ΔPredicted ET) between the 'Mod' and 'Mod + Age' models at both sites in Figure 8 (see section 6.3.3 for details). Both heatmaps demonstrate clear, yearly cyclical patterns of SHAP importance for all features apart from `P_sum_14D` and `W_speed` in 'Mod + Age'. Visual inspection of `Site_age` SHAP values appear to line up on the heatmap with the ΔPredicted ET plot shifts from the $y = 0$ baseline for both sites. For example, drops from the baseline during Autumn to Spring most years at US-UMd appear to line up consistently with lower SHAP values for `Site_age` (Figure 8a). Similarly, at DE-Lnf, high peaks above the baseline appear to temporally correspond with high SHAP values for `Site_age`, such as in late 2007, mid 2010, and early 2012 (Figure 8b).

The ΔPredicted ET plot consists of differences caused solely from the addition of age as an input feature in the RF model, however this does not mean that these changes to ET predictions are caused by the variable `Site_age` itself. RF models will adapt how predictive decisions are split on all features if a single new feature is added, and shifts in predictions between models could be caused by indirect changes to how other features, even if present in both models, are used. Therefore, while `Site_age` heatmap values appear to line up with the ΔPredicted ET plot for both sites, more rigorous testing was undertaken to assess if the two are correlated or if the `Site_age` heatmap rows are simply confounded with another variable driving these ΔPredicted ET.

Scatter plots of ΔPredicted ET against `Site_age` SHAP values for the sites (Figure 9 and Figure 10) both indicate a strong, near linear relationship. Multiple linear regression models were formed using `SW_rad`, `cos_DOY`, and `Site_age` SHAP values as predictors with ΔPredicted ET as the dependent variable. At both sites, `Site_age` SHAP values were a strong and highly significant predictor of ΔPredicted ET (US-UMd: $\beta = 2.47, SE = 0.049, p < 0.01$; DE-Lnf: $\beta = 2.49, SE = 0.036, p < 0.01$). Comparatively, `SW_rad` and `cos_DOY` SHAP values were minor, but significant predictors of ΔPredicted ET in all but `SW_rad` at US-UMd which was not significant (full results for sites shown in Table 2 and Table 3). These results strongly suggest that the changes in ET predictions between 'Mod' and 'Mod + Age' were driven primarily by the use of `Site_age` within the two RF models.

2.4.4 *Manipulating Training Ages*

To investigate whether the RF models learn any form of true age-ET relationship when training 'Mod + Age', three new models were formed: (1) 'Mod' (2) 'Mod + Age' (3) 'Mod + Shuffled Age', where training site ages were randomly shuffled between sites (see section 6.3.3 for details).

Results from this analysis are shown in Figure 11. Consistent with previous findings, daily ET predictive performance increased with the addition of age as a feature at US-UMd ($R^2_{Mod} = 0.86 \pm 0.00$; $R^2_{Mod+Age} = 0.90 \pm 0.00$), and decreased at DE-Lnf ($R^2_{Mod} = 0.90 \pm 0.00$; $R^2_{Mod+Age} = 0.86 \pm 0.00$) in models formed with all five `NumPy` seeds. At both sites, the shuffled age model moved predictive performance back towards the 'Mod' model despite the inclusion of age (US-UMd $R^2_{Shuffled} = 0.85 \pm 0.03$; DE-Lnf $R^2_{Shuffled} = 0.89 \pm 0.02$).

2.5 Results Summary

Across all analyses, site age as a feature in RF models caused site-specific and nuanced results for daily ET predictions. Significant predictive accuracy against train-test similarity relationships (age and Gower's) were reported for all medoid sites together, although these were not as robust when measured within sites. Adding age was shown to increase predictive performance at US-UMd and reduce it at DE-Lnf despite both sites displaying strong baseline predictability. SHAP analysis as a time series revealed a temporally consistent impact on ET predictions with site age included, and regression modelling these SHAP values against the change in ET predictions between 'Mod' and 'Mod + Age' strongly suggested site age was driving these predictive changes. Shuffling ages between sites during training caused predictive accuracy to shift away from the 'Mod + Age' predictions towards the 'Mod' predictions (increasing R^2 at DE-Lnf and decreasing at US-UMd), despite age being present. Together these results reveal a nuanced affect of site age as a feature in RF models which appear to be site-specific but detectable across numerous modelling approaches.

3 DISCUSSION

What this study addressed 100 words

This study used ML models to form predictions of daily ET at FLUXNET forest sites across Europe and North America. While many previous studies have used FLUXNET with ML for ET predictions [**<empty citation>**], using the age of a forest as a feature in these models has not been explored previously, and this study represents the first use of this feature for predictive ET modelling using ML methods.

Key findings/results 250 words

In all analyses contained within this study, site age contributed a meaningful impact on predictions of daily ET formed by RF models. The impact of using site age resulted in nuanced predictive accuracy changes, with large variation in the direction and magnitude of these performance shifts between sites in most analyses.

Interpret results and compare with other papers/literature 600-700 words

3.0.1 Clustering

Cluster analysis with PAM has been used successfully in hydro-ecology research previously [9, 10], but literature searches suggest that its use in ML ET predictive modelling is not common. Across cluster medoids in this study, model performance was highly diverse, with one suggestion for this being that training sites were pooled from a combination of sites across ecologically different clusters in all analyses. This may have led to under representation of specific cluster types in the training data and could explain why predictions at US-Me3, a geographically and characteristically distant site compared to others in the study (Figure 1b and Table 1), were poor across all analyses - only 3 sites exist within the defined cluster, meaning that a maximum of 2 sites in the training pool could ever pass information into the RF models for predictions. Ferreira, Cunha, and Fernandes Filho [11] avoided this issue by training specific models which were trained and tested internally within each cluster. Clustering in Ferreira, Cunha, and Fernandes Filho [11] was also defined by variables which form the base of the FAO56-PM equation (used to calculate reference evapotranspiration ET_0 [12]), ensuring that the resulting clusters are relevant to the ET estimations. While not formally verified, PAM clustering using Gower's distances in this study appeared to capture climate-relevant patterns, and could be of use for future research in ET generalisation modelling as a method to quantify regional diversity.

3.0.2 Model Sensitivity and Feature Selection

Given the limitation of ML methods to extrapolate outside of feature values which have been 'seen' during training [**<empty citation>**], input feature selection has direct implications in model accessibility, interpretability, and predictive accuracy [3]. Initial sensitivity testing at cluster medoids served two purposes. It provided direct evidence that feature selection can enhance and degrade model performance differently across sites, in agreement with the idea that no single, universally optimal feature set exists for ET predictions [3]. This testing also established that the inclusion of forest age as a feature provided a measurable impact: the addition of site age was never neutral in terms of performance in any of the medoid test site predictions, providing initial evidence that site age is impacting the model's decision structure before more in depth analyses were undertaken. Models trained using easily acquired or measured variables should improve accessibility to ET predictions in data sparse regions, but predictive accuracy may be limited compared to models which are able to learn from a greater number of input sources [3]. The most predictable medoid sites in this study - US-UMd, DE-Lnf, and CA-Obs - typically followed an upward trajectory before plateauing with an increasing number of input features, and typically a strong level of predictive accuracy could be reached using just 2-3 input features, in agreement with recent findings in Nayak et al. [13].

3.0.3 ET Generalisation

Results from sampled training datasets in section 2.3 were also highly variable in overall performance and performance of models including site age. Ahmadi et al. [14] formed models with the aim of enhance generalisation capacity and reduce overfitting predictions to a single site by training on diverse time series data from multiple sites, in a manner similar to the present study. However, model performance from Ahmadi et al. [14] was far stronger and more stable across unseen sites compared to the present study. This is likely because, as in Ferreira, Cunha, and Fernandes Filho [11], training and testing sites were confined to much smaller and similar geographic regions; all 55 sites from Ahmadi et al. [14] were contained within the Central Valley of California - a climatically homogeneous region. In turn, training and test data were likely confined within a much narrower range compared to this study, potentially enabling higher predictive accuracy - although this is also confounded with the use of deep learning and automatic hyperparameter optimisation per model in Ahmadi et al. [14], which was not utilised here. This claim is partially supported by regression analysis in section 2.3, where overall site similarity to the test site (using Gower's distance) was able to predict R^2 in both models across the full set of medoid sites, with accuracy falling as similarity reduced. This relationship was also significant at the finer site-level scale for 4/6 medoid test sites, indicating that the all site trend was not just an artifact from cross-site pooling.

3.0.4 Forest Site Age as a Feature

This study was building on work carried out in Besnard et al. [5] in which forest age was used to

Address limitations and potential impact on results 300 words

Implications and directions on/for future research 250 words

4 CONCLUSION

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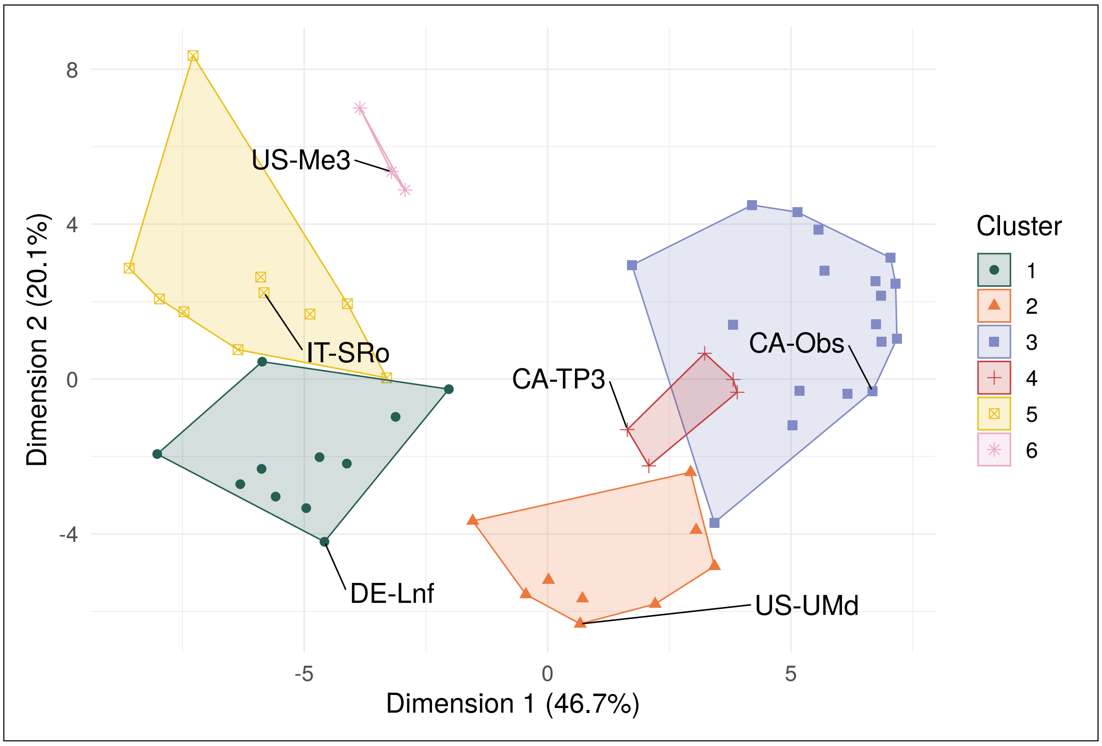
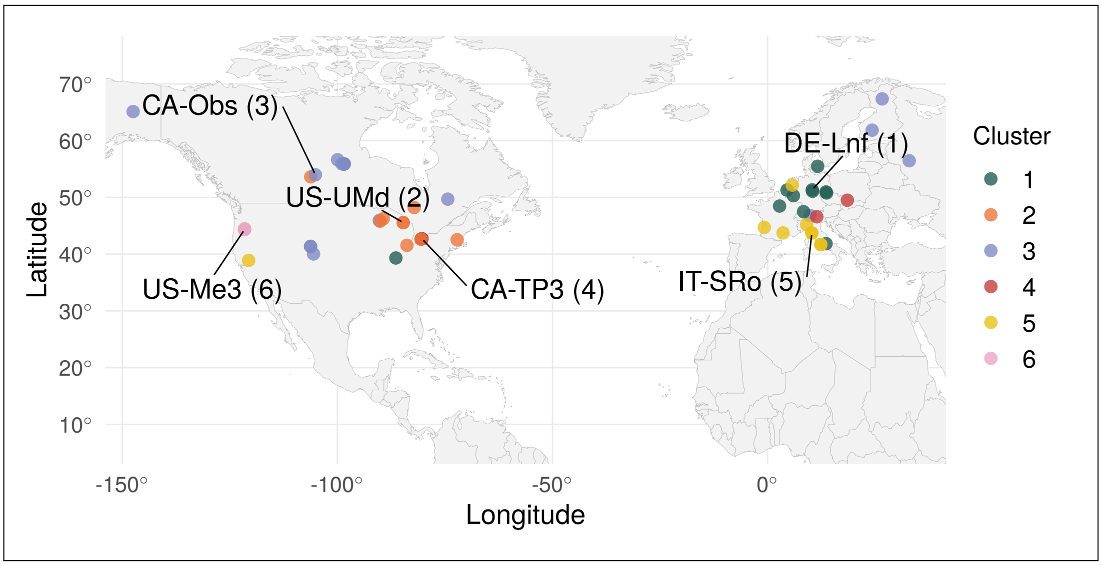
5 FIGURES AND TABLES



(a) Cluster plot indicating the site positions in the space of the first two principal components, with coloured clusters indicating the PAM solution. Axes indicate the % variance explained by each dimension.



(b) Map of all FLUXNET sites and associated clusters in this study, with cluster medoid site IDs highlighted. Visual inspection of cluster geographic distribution aligns with an expected pattern with regards to climatic groupings, suggesting the clustering solution captures climate-relevant patterns.

Figure 1: PAM solution from analysis at $k = 6$ clusters and using Gower's distances shown on a cluster plot in a, and geographic cluster distributions shown in b.

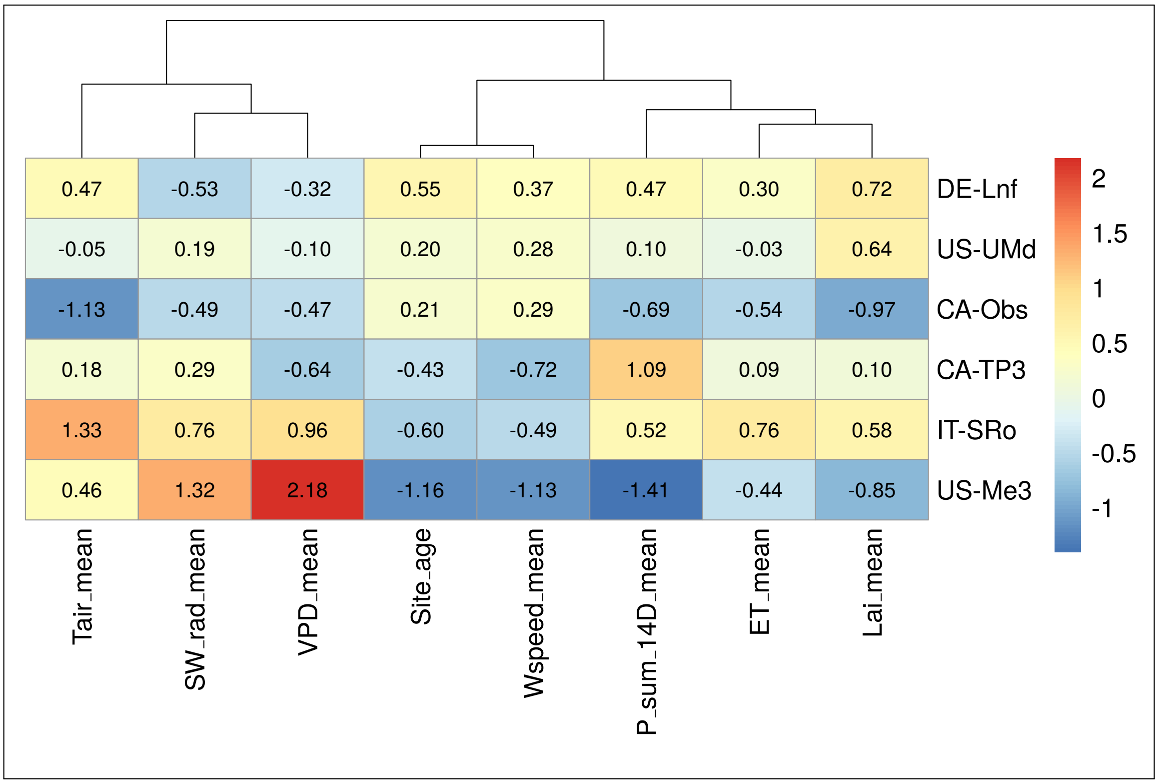


Figure 2: Heatmap of cluster medoid sites and scaled feature values. Scaled features $\geq$ 1.5 standard deviations from the mean of all sites were classed as 'very' (high or low, depending on direction), while features between 1 and 1.5 standard deviations from the mean were classed as high or low, depending on direction, in order to characterise clusters.

Table 1: Main characteristics of each cluster defined during cluster analysis

Cluster	Medoid Site	Defining Characteristics
1	DE-Lnf	Temperate sites without significant features.
2	US-UMd	Continental sites without significant features.
3	CA-Obs	Continental sites with low mean air temperatures.
4	CA-TP3	Continental sites with high precipitation.
5	IT-SRo	Temperate sites with high air temperatures.
6	US-Me3	Temperate sites with very high VPD, high SW radiation, low stand age, low wind speeds, and low precipitation.

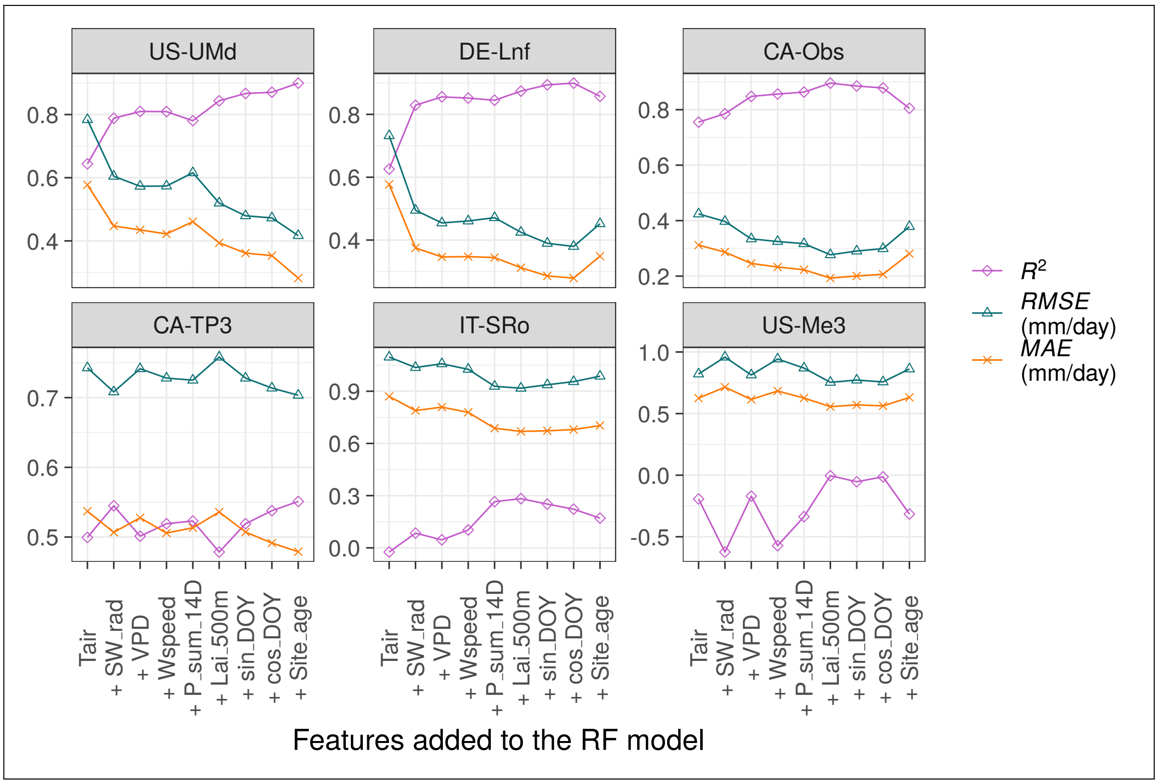


Figure 3: R^2, RMSE, and MAE for medoid cluster site predictions of daily ET during sensitivity testing (ordered left-right and top-bottom by peak R^2). Models were built and tested using a consecutively greater number of input features from FLUXNET and LAI data (for example, the '+VPD model' incorporates Tair, SW_rad, and VPD as training and input features). R^2 performance at only half of the sites was moderate or strong, and suggests that adding features up to LAI improved prediction accuracy at four of the six medoid sites. Adding site age as a final feature led to mixed performance across ET predictions made at medoid sites, with only two of the six benefiting in terms of predictive performance from this input feature.

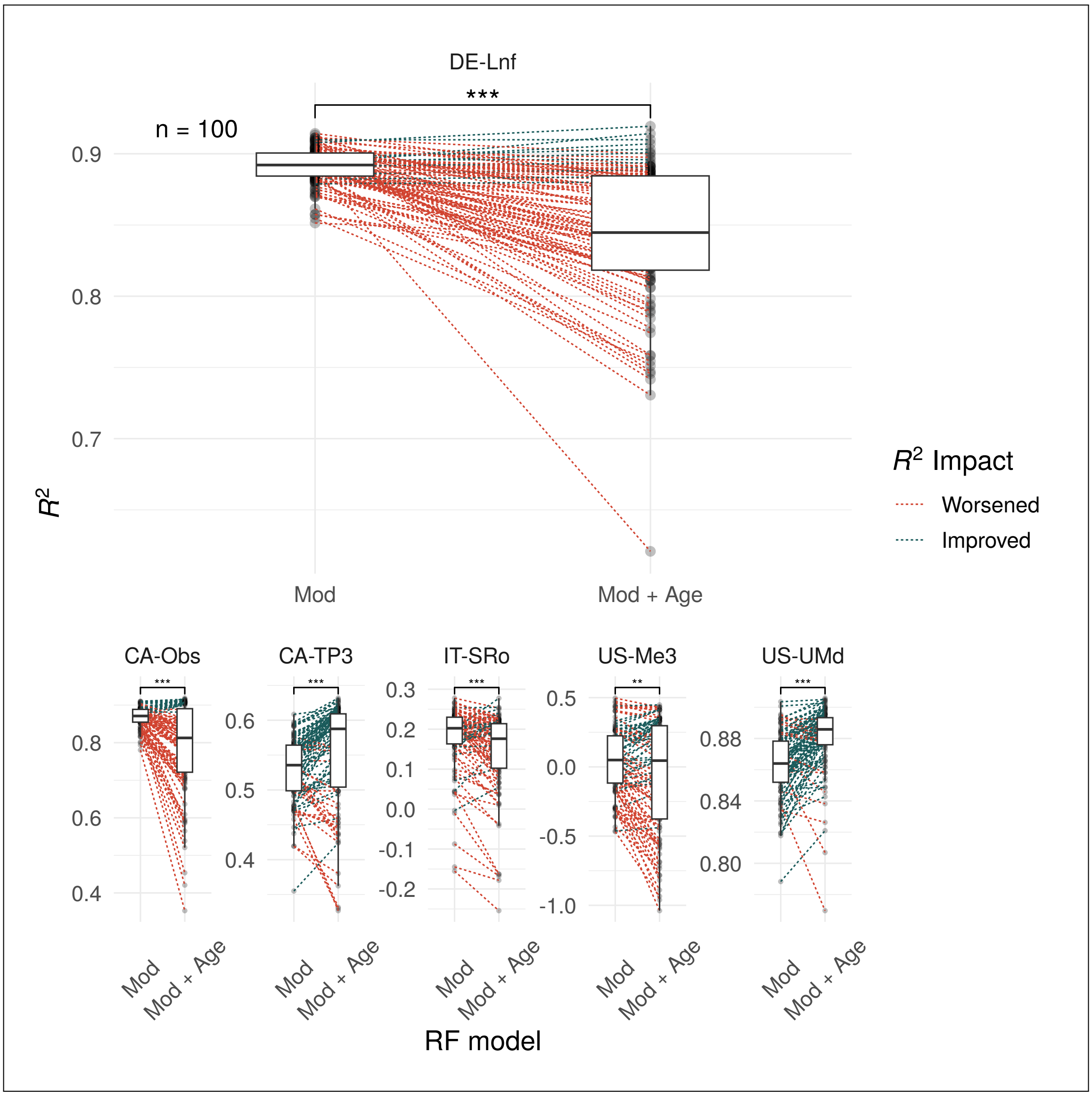


Figure 4: Paired box-plots indicate the impact on R^2 when age is added as a feature to RF models across 100 repeated runs of randomly sampled training datasets. Points are connected on a per-run basis where training dataset site composition is identical. Wilcoxon signed-rank tests suggest model performances differ significantly at all test sites, as indicated by the significance annotation on each facet (*** signifies $p < 0.001$, ** signifies $p < 0.01$ [15]). DE-Lnf is shown large for readability as an example, with other cluster medoid sites shown at a reduced scale for comparison.

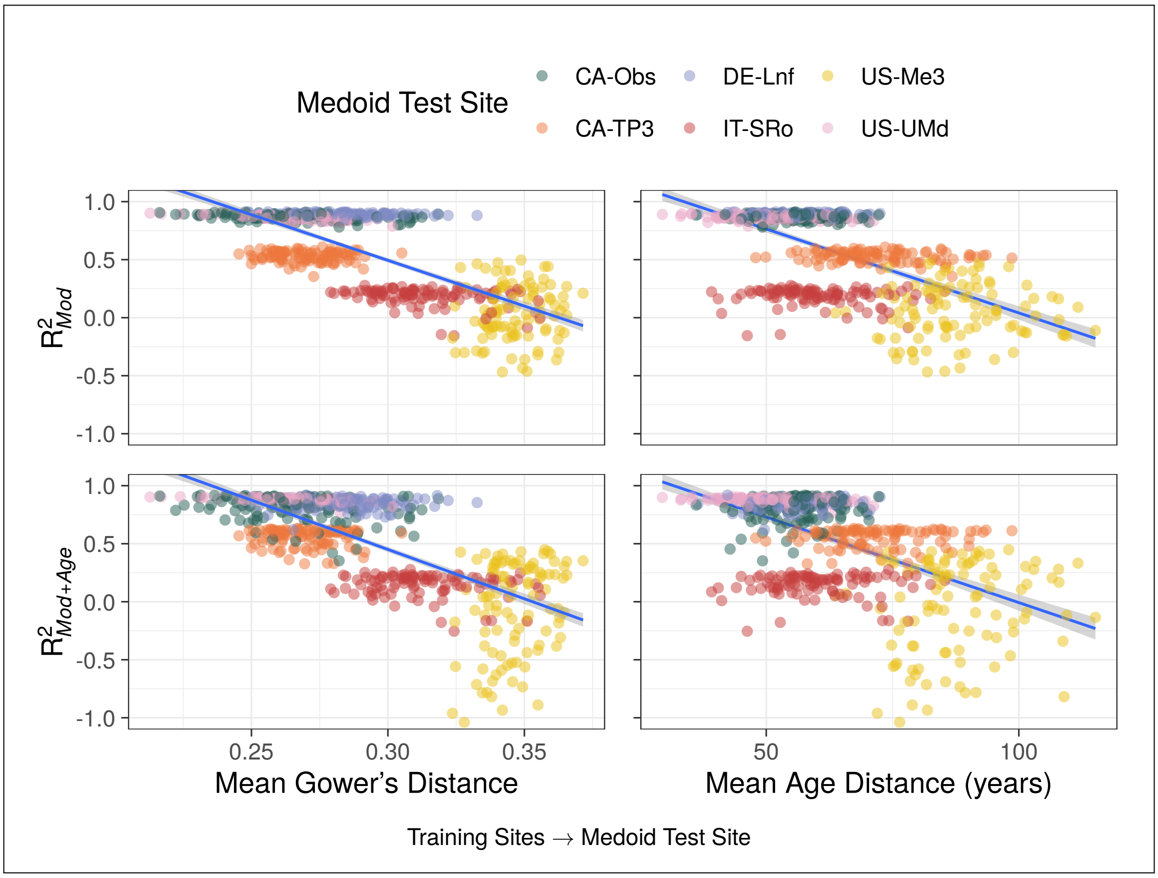
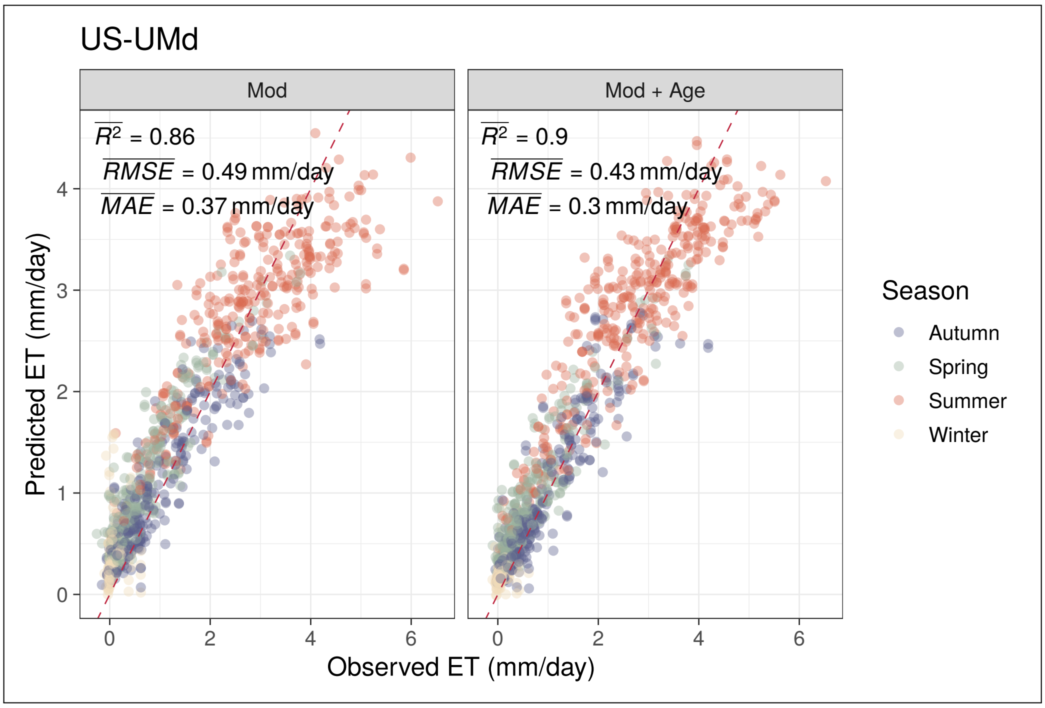
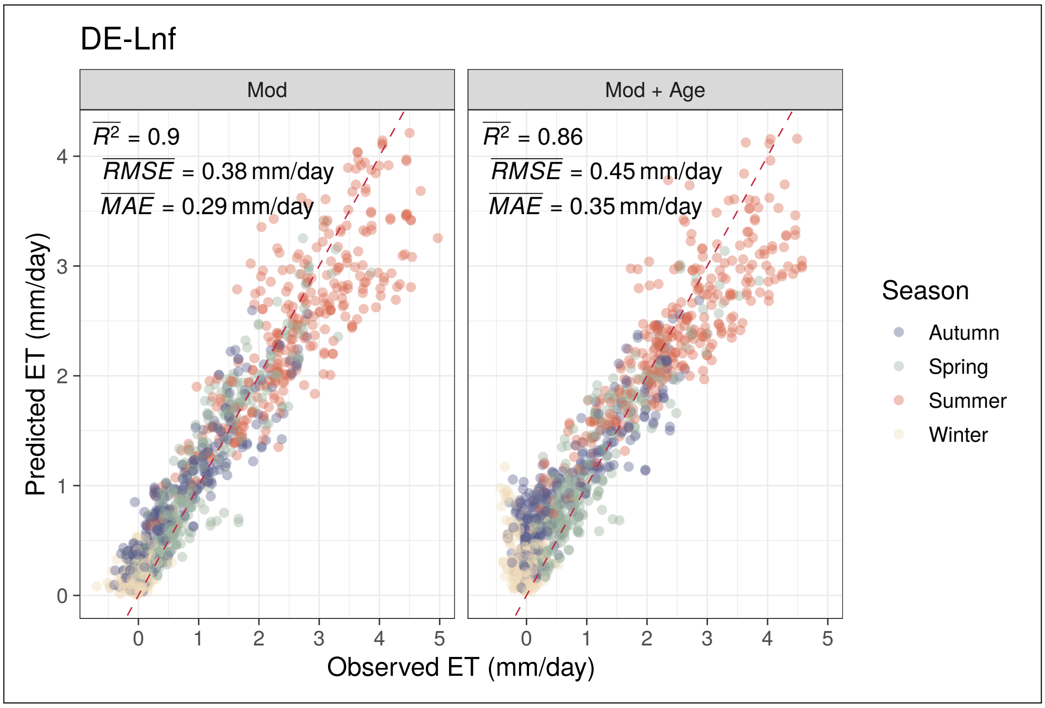


Figure 5: Plots displaying how R^2 of ET predictions relates to the mean Gower's distance and mean age distance from the training sites to each medoid test site with all sites together (100 runs). All fitted lines are a simple single linear regressions. The top plots are results from Mod, and the lower plots are results from Mod + Age, with results from all medoid test sites shown per plot. Visualisation indicate a correlation between R^2 and the mean Gower's distance in both Mod and Mod + Age, implying that models formed from datasets more closely related to the test sites produce more accurate predictions of ET. A relationship between R^2 and the mean age distance can be seen on the side plots, where predictions at test sites whose ages were further from the mean age of the training sites were typically lowered in both models.

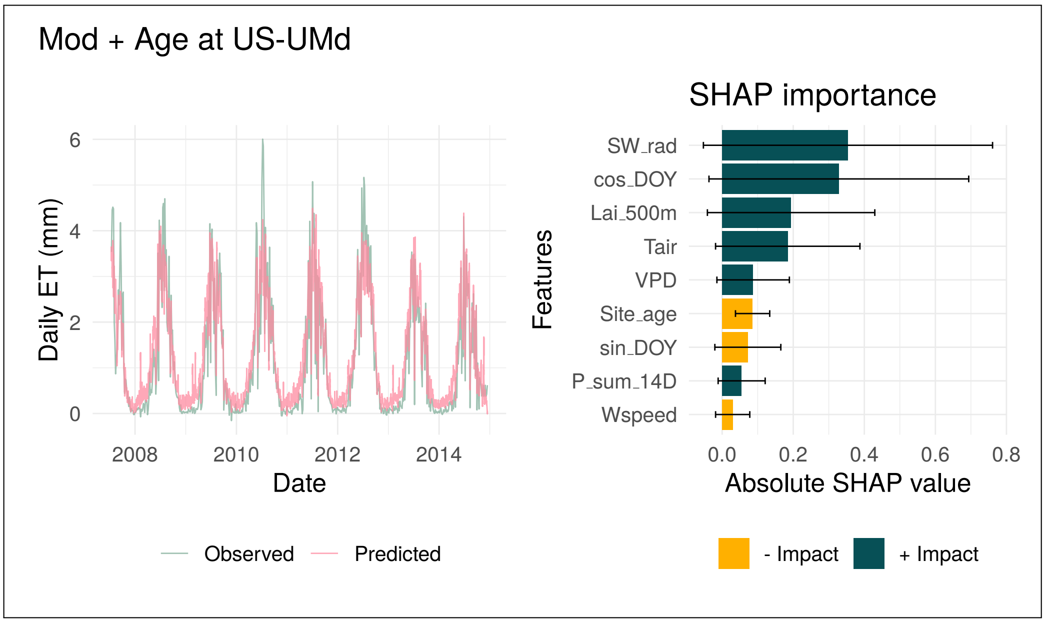


(a) Subset of 5000 points comparing the observed daily ET and predicted daily ET from the two RF models with a 1:1 line shown in red (perfect model line). 'Mod + Age' improved prediction metrics overall compared to 'Mod', with a higher R^2 and lower RMSE and MAE. Visually, points appear tighter to the 1:1 line at both extremes, although extreme highs in observed daily ET rates are often under-predicted in both models.

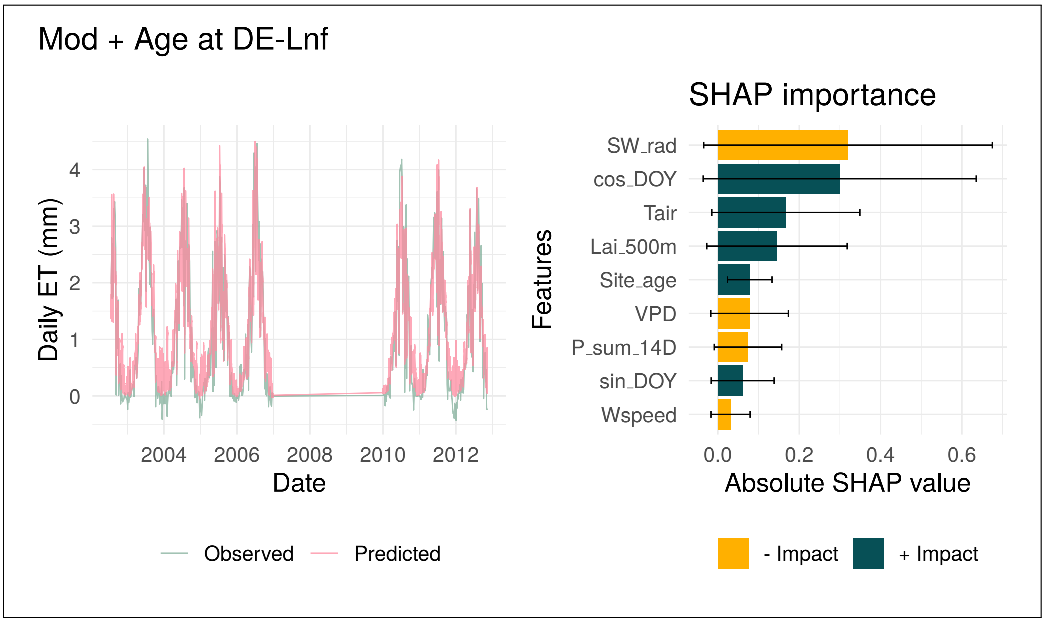


(b) Subset of 2500

Figure 6: Case Study plots for site US-UMd.

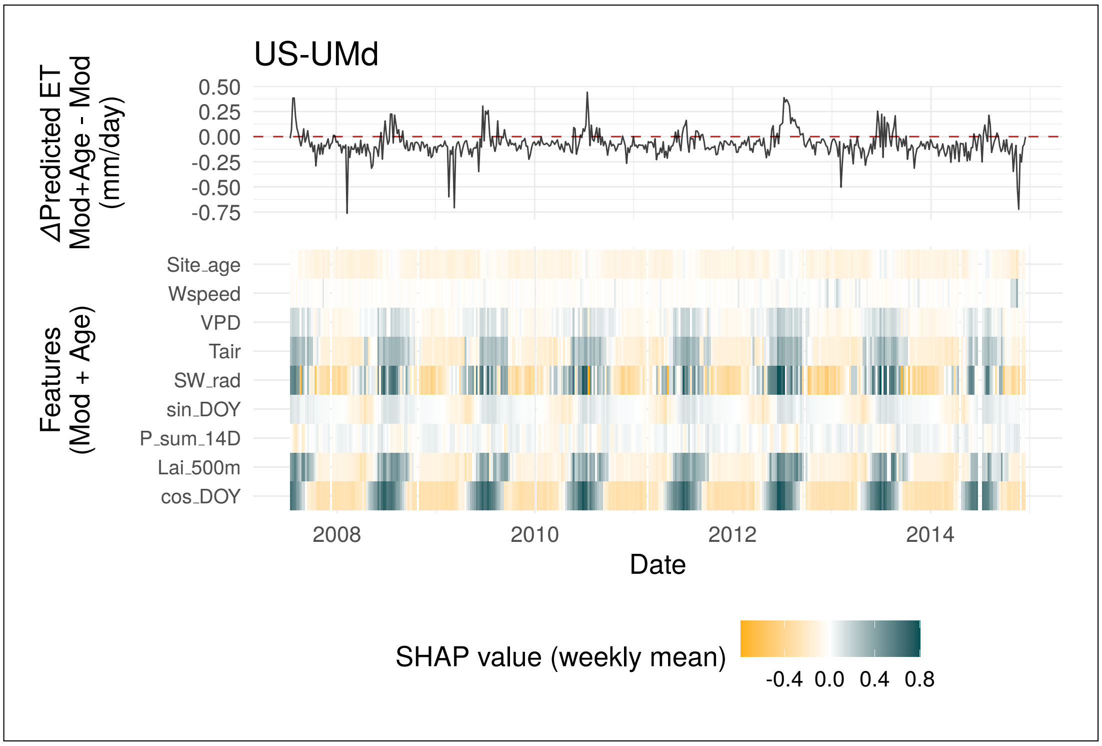


(a) Time series observed daily ET overlaid with the predicted daily ET using the 'Mod + Age' RF model seen on the left, with average SHAP feature importance values across all predictions made with this model on the right. Time series plot suggests a strong fit to the observed data but with peak ET rates in extreme summers being under-predicted, in line with the shape seen in plot a. Feature importance with SHAP...[CONTINUE]

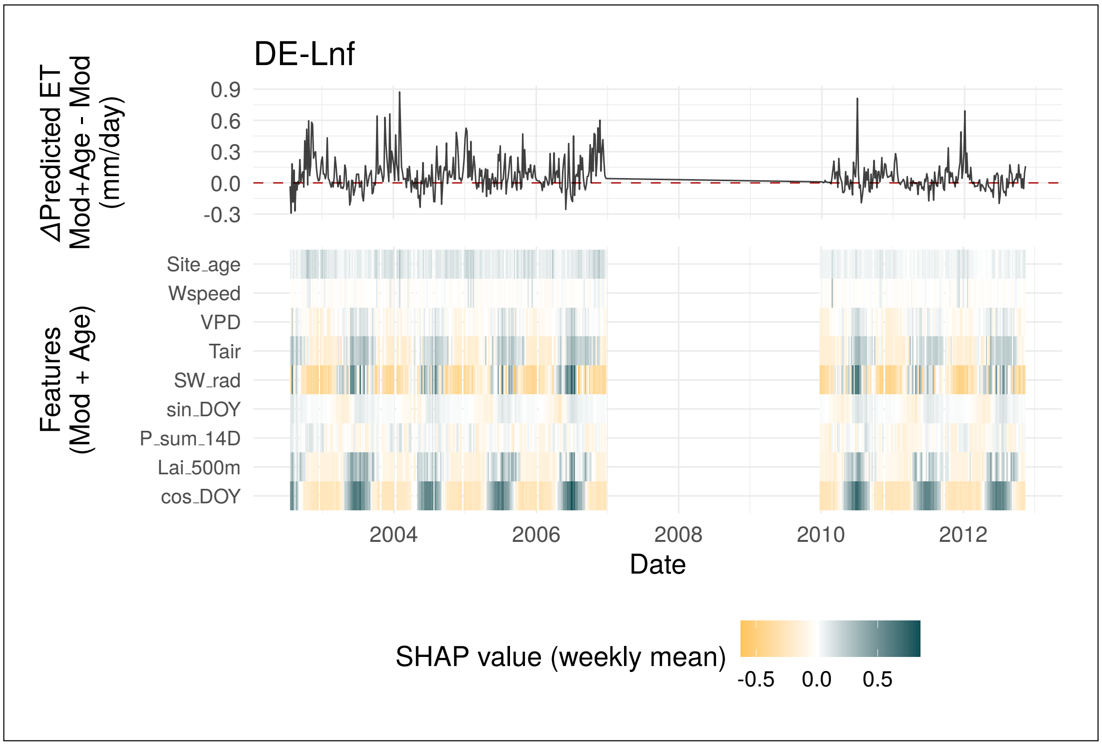


(b) M.

Figure 7: Case Study plots for site DE-Lnf.



(a)



(b) yer

Figure 8: C

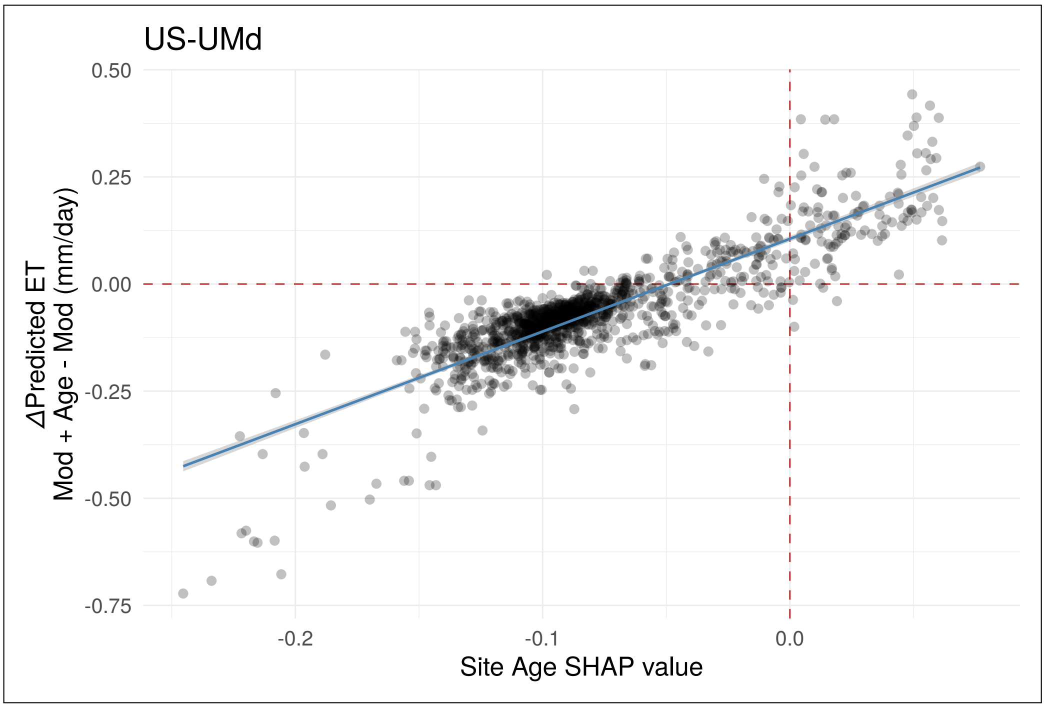

US-UMd
0.50
0.25
0.00
-0.25
-0.50
-0.75
ΔPredicted ET
Mod + Age - Mod (mm/day)
-0.2
-0.1
0.0
Site Age SHAP value

Figure 9: yh

Table 2: Multiple linear regression model results for US-UMd [16]

	Dependent variable:
	ET_diff
Site_age	2.467^{***}
	(0.049)
cos_DOY	-0.064^{***}
	(0.007)
SW_rad	0.004
	(0.005)
Constant	0.131^{***}
	(0.004)
Observations	1,250
R^2	0.766
Adjusted R^2	0.766
Residual Std. Error	0.058 (df = 1246)
F Statistic	$1{,}361.375^{***}$ (df = 3; 1246)
Note:	*p<0.1; **p<0.05; ***p<0.01

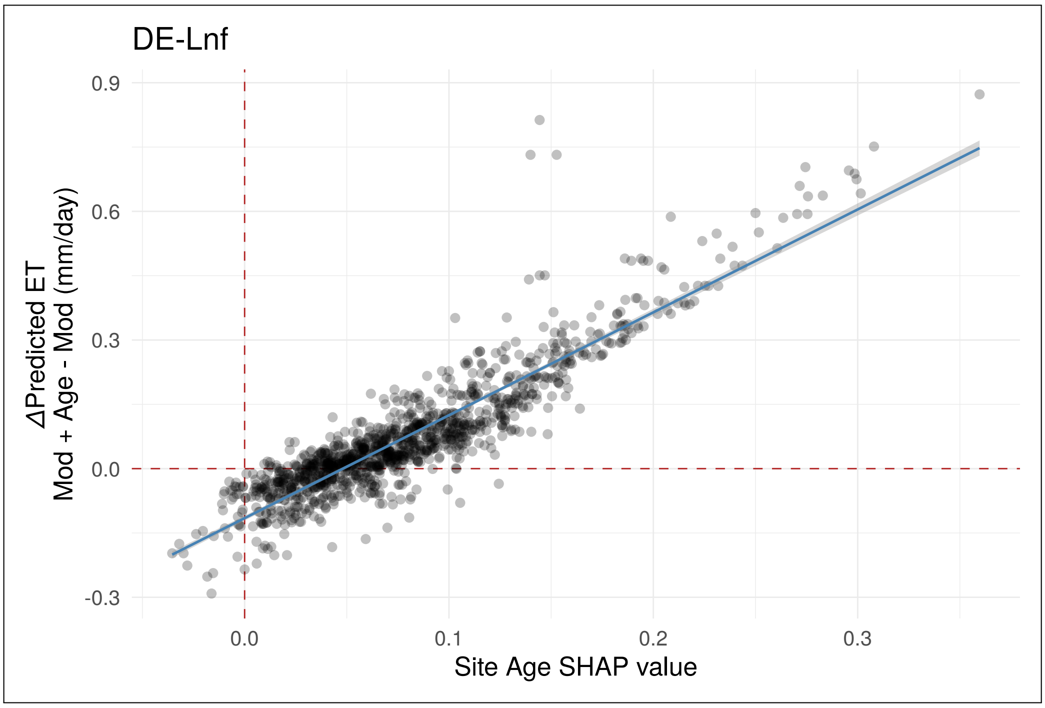


Figure 10: bbb

Table 3: Multiple linear regression model results for DE-Lnf [16]

	Dependent variable:
	ET_diff
Site_age	2.494***
	(0.036)
cos_DOY	−0.036***
	(0.006)
SW_rad	0.043***
	(0.007)
Constant	−0.117***
	(0.003)
Observations	1,250
R^2	0.830
Adjusted R^2	0.829
Residual Std. Error	0.060 (df = 1246)
F Statistic	2,025.979*** (df = 3; 1246)
Note:	*p<0.1; **p<0.05; ***p<0.01

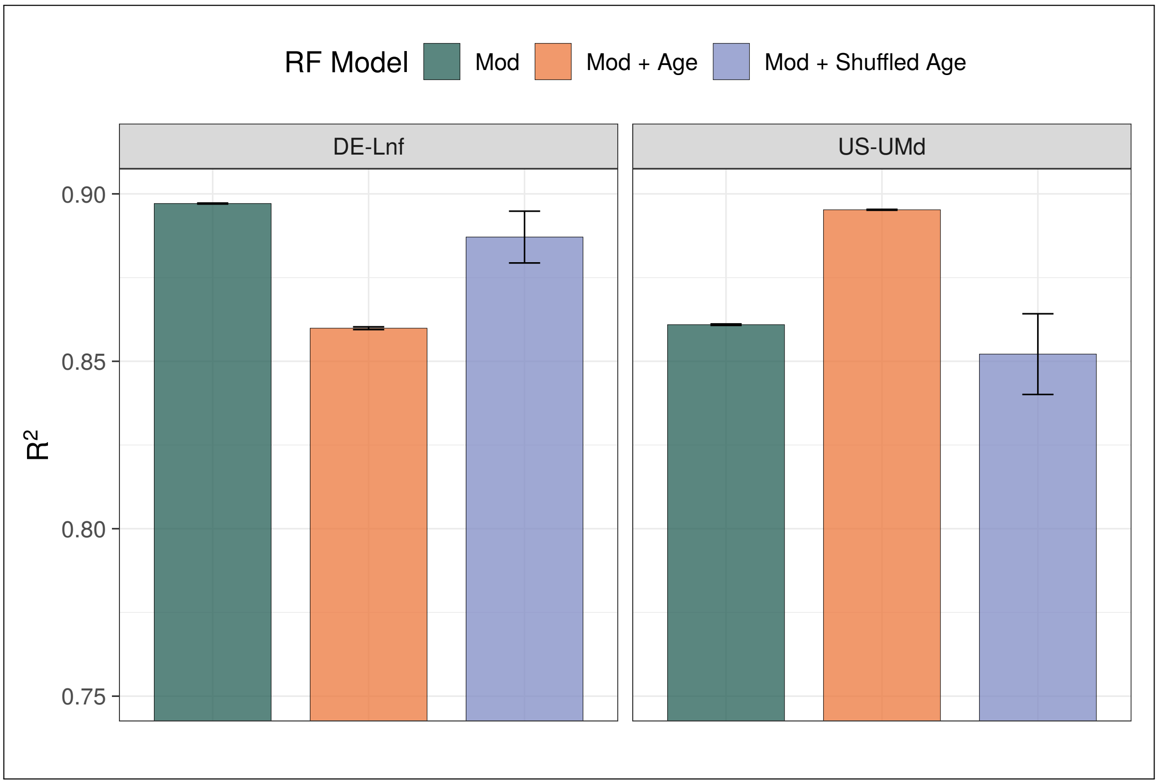


Figure 11: Bar plot showing the effect of different model setups on R^2 at DE-Lnf and US-UMd, with error bars indicating standard error. To assess whether site age was acting as a proxy for other site-level characteristics in these models, a model which randomly shuffled ages between sites was also formed using identical seeds and LOGO approach. Predictions form the 'shuffled age' model resulted in an increased R^2 at DE-Lnf compared to the 'Mod + Age' model, where an opposite effect was observed at US-UMd, consistent with previous findings between the two models. This suggests that both models are learning from true age-ET relationships

6 SUPPLEMENTARY MATERIALS AND METHODS

All scripts used in this study, alongside a guide of how to re-run the full analysis, can be found at: `https://github.com/archiebenn/forest-age-ET.git`

6.1 Data Acquisition and Variables Setup

Flux tower data were downloaded from the FLUXNET2015 dataset [17] and processed using `R` (version 4.5.3) [18] with the `tidyverse` [19] suite of packages. Flux data were filtered to only keep North American and European sites whose forest ages are dated in Besnard et al. [5], with rows of variables within these sites filtered based on their provided FLUXNET QC flag value. ET rates were calculated row-wise from the provided `LE` (latent heat flux) and `T_air` columns using the `LE.to.ET()` function from `bigleaf` [20] and multiplying this value by 86400 ($\mathrm{s\,d^{-1}}$). Therefore, ET represents actual ET (AET) throughout this study. Four-day LAI data (NASA product: MCD15A3H [21]) were retrieved using site coordinates via `MODISTools` [22], filtered on QC, and linearly interpolated between the four-day periods of LAI to provide a continuous daily value, before joining by date and site to the main FLUXNET dataset.

Site stand ages retrieved from Besnard et al. [5] are static despite multi-year data ranges often contained at these sites in the FLUXNET. As such, site ages were extrapolated across the range of data at each site to act as a continuous metric, with the provided age set to the first row of the FLUXNET data for each site. Köppen climate sites were retrieved and attached based on site latitude and longitude, and these classifications were further simplified to their main groups (Temperate and Continental in this case). To represent the day of year (DOY), $\sin()$ and $\cos()$ functions were applied to the radian value of the day within the year ($\frac{2\pi \mathrm{DOY}}{365}$) and together provide a cyclical representation of the day of year. A two week cumulative sum of precipitation prior to the day of measurement was also calculated and attached per row, with the first 13 days of data at each site then being discarded.

6.2 Cluster Analysis

Cluster analysis is an exploratory technique which aims to classify objects into as-yet-unknown groups based on their similarities [23]. To group sites in this study using cluster analysis, a Gower's distance matrix was first formed. Gower's distances are a similarity measure which can handle both numerical and categorical data to provide a distance value between two data points (sites) from 0 (completely dissimilar) to 1 (completely similar) [6]. A table of the site features used for these distance calculations can be seen in Table 4, and in this study Gower's distance is treated as a measure of ecological dissimilarity between pairs of sites. The Partitioning Around Medoids (PAM) algorithm and silhouette method was then used to map the Gower's distance matrix into a specified set of clusters [7]. PAM identifies medoids, which are the points within each cluster whose sum of dissimilarities to all other sites in the same cluster is minimal, and so, in this case, can be identified as the most representative flux tower sites of each cluster.

Table 4: Site-level variables and definitions used to calculate Gower's distances during cluster analysis.

Variable	Data Type	Definition
`ET`	Numerical	Mean daily ET rate ($mm\,d^{-1}$)
`T_air`	Numerical	Mean air temperature at site (°C)
`SW_rad`	Numerical	Mean daily incoming short-wave radiation ($W\,m^{-2}$)
`VPD`	Numerical	Mean daily vapour pressure deficit (hPa)
`Site_age`	Numerical	Mean forest stand age at site (years)
`Wspeed`	Numerical	Mean daily wind speed ($m\,s^{-1}$)
`P_sum_14D`	Numerical	Mean two-week cumulative sum of precipitation prior to the day of measurement (mm)
`Lai_500m`	Numerical	Mean daily leaf area index ($m^2\,m^{-2}$)
`Cover_type`	Categorical	IGBP cover type classification of site
`Climate_zone`	Categorical	Main Köppen climate group of site

6.3 Random Forest Modelling and Evaluation Metrics

Random Forest models were trained and tested with `RandomForestRegressor()` from `scikit-learn` [24] using the different methods outlined below. Due to time constraints, a single set of optimised parameter values was calculated once using `GroupKFold()` and `Optuna` [25] to minimise root mean square error (RMSE) on fold predictions. These values were used for all RF models throughout the study (see Table 5).

In all cases, regression analysis metrics of R^2 (coefficient of determination), RMSE, and mean absolute error (MAE) were calculated from model predictions of ET using their respective `scikit-learn` functions alongside observed values of ET at each site. R^2 aims to explain the proportion of variance captured by the model, while RMSE emphasises the size of errors against observed values. MAE was also used due to its simplicity and ability to compare to RMSE for a greater understanding of error structure. While no consensus has been reached on a unified metric to evaluate ML models and predictive accuracy, a model with the the highest values of R^2 and lowest values of RMSE is deemed the most accurate here [26, 3].

Table 5: Parameters selected for use in all RF training setups in this study. Calculated to minimise RMSE on trials using `GroupKFold()` and `Optuna`.

RF parameter	Value
n_estimators	350
max_depth	17
min_samples_leaf	19
max_features	0.35

Table 6: All Features and definitions included in different RF model setups during training and inference.

Variable	Definition
ET	Mean daily ET rate ($mm\,d^{-1}$)
Site_age	Mean forest stand age at site (years)
T_air	Mean air temperature at site (°C)
SW_rad	Mean daily incoming short-wave radiation ($W\,m^{-2}$)
VPD	Mean daily vapour pressure deficit (hPa)
Wspeed	Mean daily wind speed ($m\,s^{-1}$)
P_sum_14D	Mean two-week cumulative sum of precipitation prior to the day of measurement (mm)
Lai_500m	Mean daily leaf area index ($m^2\,m^{-2}$)
sin_DOY	Cyclical day of year representation alongside cos_DOY
cos_DOY	Cyclical day of year representation alongside sin_DOY

6.3.1 *Model Sensitivity Testing*

To assess the sensitivity of the RF models to different features, a forward feature selection assessment was carried out. Non-target (ET) numerical features were added sequentially and the RF model was re-trained and tested by forming daily ET predictions at the medoid sites with each new feature added. Features were added in the following order: `T_air`, `SW_rad`, `VPD`, `Wspeed`, `P_sum_14D`, `Lai_500m`, `sin_DOY`, `cos_DOY`, `Site_Age`. Training sites consisted of all sites apart from medoids (see Table 7), with predictions formed and testing carried out at medoid sites.

6.3.2 *Random Sampling of Training Datasets*

To investigate how predictive performance is impacted by including site age as a feature, two separate RF architectures were formed: 'Mod' (Equation 1) and 'Mod + Age' (Equation 2).

$$ET_i = RF\left(T\_air_i, SW\_rad_i, VPD_i, W\_speed_i, \sum_{i-13}^{i} P, LAI_i, \sin\_DOY_i, \cos\_DOY_i\right) \quad (1)$$

$$ET_i = RF\left(T\_air_i, SW\_rad_i, VPD_i, W\_speed_i, \sum_{i-13}^{i} P, LAI_i, \sin\_DOY_i, \cos\_DOY_i, Site\_Age_i\right) \quad (2)$$

where the target ET_i is the estimated ET rate on the ith day (mm/day); $RF(\cdot)$ is the Random Forest model; the features T_air, SW_rad, VPD, W_speed, $\sum_{i-13}^{i} P_i$, LAI, $Site\_Age$ are all listed with descriptions in Table 6.

To further assess how the overall ecological similarity of training dataset composition to the test site may impact predictive accuracy of daily ET, a method to subset sites from a larger site pool before RF training was also developed allowing Gower's distances of training sites to be varied and potential effects investigated (script `21_analysis_1.2.py`), with details on how this analysis runs below.

Sites were first separated as follows: 6 medoid sites for testing and the remaining 44 sites as the full training pool. Following this, 20 sites were chosen at random (`NumPy.random` module) from the training pool, and then two models - 'Mod' and 'Mod + Age' - are trained on this subset of sites. After training, both models then predict on the six held out medoid sites, and the mean Gower's distance and mean age distance from the full training subset to each test site is calculated and stored alongside prediction performance metrics. This paired process is then repeated a further 99 times, each with a new `NumPy` seed to avoid using the same subset, to produce 100 runs using this method.

this is where i need to explain gower's distance mean plot etc.

6.3.3 *Two Site Case Study: US-UMd and DE-Lnf*

- details on LOGO approach - random shuffle details - heatmap and ΔPredicted ET details

6.4 Site Metadata

Table 7: Metadata summary of FLUXNET sites used in this study (*** denotes a cluster medoid)

FLUXNET ID	Days of Data	Country	Climate Zone	Cover Type	Site Age (years)	Mean ET 90D Peak (mm d^{-1})
BE-Bra	3590	Belgium	Temperate	MF	92	2.02
BE-Vie	4466	Belgium	Temperate	MF	107	2.15
CA-Gro	3639	Canada	Continental	MF	84	2.45
CA-Man	2290	Canada	Continental	ENF	173	1.90
CA-NS1	1141	Canada	Continental	ENF	157	1.59
CA-NS2	960	Canada	Continental	ENF	76	1.75
CA-NS3	1141	Canada	Continental	ENF	42	1.70
CA-NS5	1137	Canada	Continental	ENF	26	1.88
CA-NS6	1141	Canada	Continental	OSH	17	1.35
CA-NS7	1141	Canada	Continental	OSH	8	1.73
CA-Oas	3029	Canada	Continental	DBF	91	2.82
CA-Obs***	3029	Canada	Continental	ENF	122	1.95
CA-Qfo	2687	Canada	Continental	ENF	106	1.75
CA-TP3***	4550	Canada	Continental	ENF	44	2.45
CA-TPD	1082	Canada	Continental	DBF	100	2.31
CH-Lae	3782	Switzerland	Temperate	MF	190	4.31
CZ-BK1	3920	Czechia	Continental	ENF	37	1.73
DE-Hai	2720	Germany	Temperate	DBF	260	2.68
DE-Lnf***	2670	Germany	Temperate	DBF	122	2.60
DE-Obe	2463	Germany	Temperate	ENF	79	2.21
DE-Tha	4502	Germany	Temperate	ENF	131	2.14
DK-Sor	4482	Denmark	Temperate	DBF	98	2.87
FI-Hyy	4466	Finland	Continental	ENF	59	2.15
FI-Sod	4454	Finland	Continental	ENF	91	1.87
FR-Fon	3586	France	Temperate	DBF	166	3.64
FR-LBr	2346	France	Temperate	ENF	44	2.76
FR-Pue	4550	France	Temperate	EBF	73	1.86
IT-Col	2908	Italy	Temperate	DBF	195	2.58
IT-Cp2	1082	Italy	Temperate	EBF	65	2.38
IT-Cpz	2359	Italy	Temperate	EBF	65	2.02
IT-PT1	897	Italy	Temperate	DBF	15	3.51
IT-Ren	3253	Italy	Continental	ENF	199	3.53
IT-SR2	716	Italy	Temperate	ENF	65	2.21
IT-SRo***	3819	Italy	Temperate	ENF	63	2.51
NL-Loo	4498	Netherlands	Temperate	ENF	119	2.64
RU-Fyo	4458	Russia	Continental	ENF	247	2.51
US-Blo	1901	U.S.A	Temperate	ENF	21	4.04
US-GBT	898	U.S.A	Continental	ENF	181	2.66
US-GLE	3638	U.S.A	Continental	ENF	190	2.36
US-Ha1	3262	U.S.A	Continental	DBF	112	2.40
US-MMS	4538	U.S.A	Temperate	DBF	105	3.23
US-Me3***	2012	U.S.A	Temperate	ENF	23	1.91
US-NR1	4550	U.S.A	Continental	ENF	121	2.71
US-Oho	3615	U.S.A	Continental	DBF	55	4.02
US-PFa	4522	U.S.A	Continental	MF	164	2.35
US-Prr	1355	U.S.A	Continental	ENF	101	1.56
US-Syv	2723	U.S.A	Continental	MF	307	2.52
US-UMB	4538	U.S.A	Continental	DBF	102	3.16
US-UMd***	2715	U.S.A	Continental	DBF	94	2.91
US-WCr	2931	U.S.A	Continental	DBF	106	2.56